

Exact Sign-Sector Probabilities for the Diagonal of a Haar-Random Rotation in $\text{SO}(3)$

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Abstract

Let R be Haar distributed on $\text{SO}(3)$. The support of $\text{diag } R = (R_{11}, R_{22}, R_{33})$ is the classical rotation tetrahedron. We derive the Haar pushforward density on this tetrahedron and compute the exact probabilities of the eight coordinate sign sectors. Every sign pattern with an even number of negative entries has probability $5/24$, whereas every sign pattern with an odd number has probability $1/24$; hence the two parity classes have total probabilities $5/6$ and $1/6$. The main proof uses unit-quaternion Hopf coordinates. A second derivation based on independent Gaussian squares provides an analytic cross-check. The result shows explicitly that the geometry of the tetrahedral support alone does not determine the probabilities induced by Haar-uniform physical rotations.

Keywords: Haar measure; random rotation; rotation matrix; diagonal entries; sign probability; Dirichlet distribution.

1 Setting and pushforward density

The possible diagonals of three-dimensional rotation matrices form a tetrahedron [1, 3]. Random rotations and Haar measure on $\text{SO}(3)$ are classical subjects [2, 4, 5]. The question considered here is the Haar probability assigned to each of the eight regions cut out by the coordinate planes inside that tetrahedron.

Represent a Haar-uniform rotation by a unit quaternion $q = (q_0, q_1, q_2, q_3)$ uniform on S^3 , and set $p_j = q_j^2$. Then

$$(p_0, p_1, p_2, p_3) \sim \text{Dir}\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right). \quad (1)$$

The diagonal entries are

$$R_{11} = p_0 + p_1 - p_2 - p_3, \quad (2)$$

$$R_{22} = p_0 - p_1 + p_2 - p_3, \quad (3)$$

$$R_{33} = p_0 - p_1 - p_2 + p_3. \quad (4)$$

Writing $r_i = R_{ii}$ and

$$L_0 = 1 + r_1 + r_2 + r_3, \quad L_1 = 1 + r_1 - r_2 - r_3, \quad (5)$$

$$L_2 = 1 - r_1 + r_2 - r_3, \quad L_3 = 1 - r_1 - r_2 + r_3, \quad (6)$$

one has $p_j = L_j/4$. Thus $L_j \geq 0$ are precisely the inequalities defining the rotation tetrahedron.

Theorem 1 (Haar density of the rotation diagonal). *Inside the rotation tetrahedron, the density of $r = (R_{11}, R_{22}, R_{33})$ with respect to Euclidean volume is*

$$f_H(r) = \frac{1}{\pi^2 \sqrt{L_0 L_1 L_2 L_3}}, \quad (7)$$

and it is zero outside.

Proof. The Dirichlet density on the simplex is $\pi^{-2} \prod_{j=0}^3 p_j^{-1/2}$. Using (p_0, p_1, p_2) as independent coordinates, the affine map $p \mapsto r$ has absolute Jacobian 16. Since $p_j = L_j/4$, the factor 16 from $\prod_j p_j^{-1/2}$ cancels the Jacobian. $\square$

The boundary singularity makes clear why uniform Euclidean volume in the projected tetrahedron is not the same probability model as Haar-uniform rotation.

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2 Exact sign-sector probabilities

The coordinate planes have probability zero. Left multiplication by a coordinate-axis half-turn flips any chosen pair of diagonal signs and preserves Haar measure. Consequently, the four even-negative patterns have a common probability a , and the four odd-negative patterns have a common probability b .

Theorem 2 (Exact Haar sign-sector probabilities). *For $s = (s_1, s_2, s_3) \in \{-1, +1\}^3$,*

$$\mathbb{P}(\text{sgn}(R_{11}, R_{22}, R_{33}) = s) = \begin{cases} 5/24, & s_1 s_2 s_3 = +1, \\ 1/24, & s_1 s_2 s_3 = -1. \end{cases} \quad (8)$$

Proof. It is enough to compute $a = \mathbb{P}(R_{11} > 0, R_{22} > 0, R_{33} > 0)$. Use Hopf coordinates

$$q_0 = \cos \eta \cos \alpha, \quad q_1 = \cos \eta \sin \alpha, \quad (9)$$

$$q_2 = \sin \eta \cos \beta, \quad q_3 = \sin \eta \sin \beta, \quad (10)$$

where $u = \sin^2 \eta$ is uniform on $[0, 1]$, while α and β are independent uniform angles. Then

$$R_{11} = 1 - 2u, \quad (11)$$

$$R_{22} = (1 - u) \cos 2\alpha + u \cos 2\beta, \quad (12)$$

$$R_{33} = (1 - u) \cos 2\alpha - u \cos 2\beta. \quad (13)$$

The all-positive event is equivalent to $u < 1/2$, $\cos 2\alpha > 0$, and $u|\cos 2\beta| < (1 - u) \cos 2\alpha$. Conditional on $\cos 2\alpha > 0$, put $X = \cos 2\alpha$ and $Y = |\cos 2\beta|$. Then X, Y are independent arcsine variables on $[0, 1]$, so

$$a = \frac{1}{2} \mathbb{E} \min\left(\frac{1}{2}, \frac{X}{X + Y}\right). \quad (14)$$

Exchange symmetry gives

$$\mathbb{E} \min\left(\frac{1}{2}, \frac{X}{X + Y}\right) = \frac{1}{4} + \frac{1}{2} \mathbb{E} \frac{\min(X, Y)}{X + Y}. \quad (15)$$

Writing $X = \sin A$ and $Y = \sin B$, with A, B independent uniform on $[0, \pi/2]$, yields

$$\mathbb{E} \frac{\min(X, Y)}{X + Y} = \frac{8}{\pi^2} \int_{0 \leq A \leq B \leq \pi/2} \frac{\sin A}{\sin A + \sin B} dA dB = \frac{1}{3}. \quad (16)$$

Appendix A evaluates the integral as $\pi^2/24$. Therefore

$$a = \frac{1}{2} \left(\frac{1}{4} + \frac{1}{6} \right) = \frac{5}{24}. \quad (17)$$

Finally, $4a + 4b = 1$ gives $b = 1/24$. $\square$

Corollary 1. *The four even-negative sectors have total Haar probability $5/6$, and the four odd-negative sectors have total Haar probability $1/6$.*

Negative-sign parity	Probability per sector	Sign patterns
Even	5/24	+++ , +- -, -+- , --+
Odd	1/24	-++ , +-+ , ++- , ---

3 Independent Gaussian-square derivation

Let g_0, g_1, g_2, g_3 be independent standard normal variables and $Z_i = g_i^2$. Normalizing the Gaussian vector produces a uniform unit quaternion, and the common normalization does not affect signs. Define

$$A = Z_0 + Z_1, \quad B = Z_2 + Z_3. \quad (18)$$

The variables A, B are independent chi-square variables with two degrees of freedom, hence independent exponentials, and $R_{11} > 0$ is the event $A > B$. Within the two pairs, let

$$U = \frac{Z_0}{A}, \quad V = \frac{Z_2}{B}. \quad (19)$$

Then U, V are independent Beta(1/2, 1/2) variables, independent of A, B . Conditional on $A > B$, the ratio $r = B/A \in [0, 1]$ has density $2/(1+r)^2$. Put $W = |2U - 1|$ and $Y = |2V - 1|$. The remaining two positivity inequalities require $2U - 1 > rY$, and therefore

$$\mathbb{P}(R_{11}, R_{22}, R_{33} > 0) = \frac{1}{4} \mathbb{E} \int_0^{\min(1, W/Y)} \frac{2 dr}{(1+r)^2} \quad (20)$$

$$= \frac{1}{4} \left[\frac{1}{2} + \mathbb{E} \frac{\min(W, Y)}{W + Y} \right] = \frac{5}{24}. \quad (21)$$

This gives the same value through independent Gaussian pair sums rather than Hopf coordinates.

4 Concluding remarks and priority boundary

The classical support tetrahedron does not determine the Haar probabilities of its coordinate sign sectors. The pushforward density is boundary-singular, and the exact sector weights split sharply into 5/24 and 1/24.

The tetrahedral support, quaternion representation, Haar generation, and random-rotation background are established. In the literature search performed for this manuscript, we did not locate the explicit values 5/24 and 1/24 stated as these diagonal sign-sector probabilities. This sentence reports the coverage of the search; it is not an exhaustive claim of historical priority. Full-text checking of the closest older sources remains ongoing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT to assist with manuscript organization, language editing, literature tracking, and independent algebraic and numerical checks. The author reviewed and verified all derivations, references, and statements, edited the content as needed, and takes full responsibility for the content of the publication.

A Evaluation of the critical integral

Let

$$J = \int_{0 \leq A \leq B \leq \pi/2} \frac{\sin A}{\sin A + \sin B} dA dB. \quad (22)$$

Set $s = (A + B)/2$ and $t = (B - A)/2$. The domain becomes $0 \leq t \leq \pi/4$ and $t \leq s \leq \pi/2 - t$, the Jacobian is 2, and

$$\frac{2 \sin(s - t)}{\sin(s - t) + \sin(s + t)} = 1 - \tan t \cot s. \quad (23)$$

Hence

$$J = \int_0^{\pi/4} \int_t^{\pi/2-t} (1 - \tan t \cot s) ds dt \quad (24)$$

$$= \frac{\pi^2}{16} - \int_0^{\pi/4} \tan t \ln(\cot t) dt. \quad (25)$$

With $x = \tan t$ and then $y = x^2$,

$$\int_0^{\pi/4} \tan t \ln(\cot t) dt = -\frac{1}{4} \int_0^1 \frac{\ln y}{1+y} dy = \frac{\pi^2}{48}, \quad (26)$$

where the last equality follows by expanding $(1+y)^{-1}$ into its alternating series. Therefore $J = \pi^2/24$.

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